An Adaptive GRPO–MaxRL Bridge for Reasoning Optimization

Anonymous submission

Abstract

Reinforcement learning is an effective way to improve language-model reasoning from outcome feedback. Widely used group-relative methods remove the critic and construct a stable centered advantage from within-group rewards, but remain local reward-linear proxies rather than approaching maximum likelihood as rollout compute grows. Maximum Likelihood Reinforcement Learning closes this objective gap through a compute-indexed objective, yet its update can become overly sharp when high-reward trajectories are sparse. Recent listwise projection methods provide another way to control update concentration, but typically require a prescribed temperature and projection strategy whose best choices can vary across tasks (Qu et al. 2026). We propose BridgeRL, which balances reward gain against effective sample support to select a tilted target and then uses retained target support to calibrate a convex interpolation between conservative group-relative and likelihood-seeking updates. Across six in-domain benchmarks, BridgeRL reaches aggregate scores of 47.1 at 9B and 40.8 at 4B. It surpasses the strongest aggregate baseline at each scale—MixRL at 43.4 and GRPO at 37.1—by 3.7 points at both scales. Relative to GRPO, it improves mathematical reasoning by 4.8 and 3.8 points and multi-hop question answering by 4.0 and 3.6 points at 9B and 4B, respectively. Additional sweeps show that fixed projection temperatures and fixed interpolation weights favor different tasks and sampling budgets, whereas the adaptive rule retains the strongest aggregate performance.

1 Introduction

Reinforcement learning (RL) has become central to improving language-model reasoning by optimizing sampled solutions from outcome feedback (Shao et al. 2024; Guo et al. 2025). Group Relative Policy Optimization (GRPO) is a particularly effective critic-free solution (Shao et al. 2024). For each prompt, it compares sampled responses using their reward relative to the group mean and variation. This centered signal is practical and stable, and it automatically changes relative emphasis according to the observed reward profile. Its behavior is nevertheless fixed by the standardization rule. GRPO remains a local reward-linear proxy rather than moving toward the maximum-likelihood objective, which directly learns to assign probability to verified successful responses and represents the desired global training objective in the idealized supervised setting. Sampling more responses can improve the estimate of the GRPO update, but it neither closes this objective gap nor allows the optimizer to choose how sharply to focus on the best responses.

Maximum Likelihood Reinforcement Learning (MaxRL) develops a different, compute-indexed objective (Tajwar et al. 2026). It uses additional rollout compute to incorporate progressively higher-order reward information and moves toward likelihood-based optimization. This makes MaxRL more aggressive in promoting the best observed trajectories and changes the objective being approximated as compute grows. The same property creates its main practical challenge: when high-reward trajectories are sparse, the update can become excessively sharp, rely on very few samples, and induce unstable or overly concentrated policy changes. This sharpness issue, rather than reward granularity, motivates an adaptive intermediate regime. MaxRL naturally admits extensions to multi-level rewards.

Recent work has also recast group optimization as listwise target projection over sampled responses (Qu et al. 2026). This perspective offers an explicit mechanism for controlling target concentration, but it introduces a temperature hyperparameter together with a choice of forward or reverse projection. Fixing the temperature applies one concentration rule across groups even though their reward gaps and high-reward support differ; empirical behavior can also depend on the projection direction. This leaves open how to determine update sharpness directly from the reward evidence available in each group.

These complementary properties motivate a continuous, group-adaptive bridge. We propose BridgeRL, which first evaluates how strongly each sampled group supports reward-seeking reweighting by balancing expected reward improvement against retained sample support. It then measures the concentration of the selected target relative to the sharpest target supported by the group and uses this normalized support to interpolate between GRPO and MaxRL. This separates target search from coefficient calibration: inverse temperature determines the candidate target, while the target's simplex concentration determines the actor update.

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We compare against GRPO, MaxRL, a fixed midpoint interpolation, and the forward and backward LPO variants. Across the six in-domain benchmarks and all reported metrics, BridgeRL reaches 47.1 at 9B and 40.8 at 4B. These results exceed the strongest aggregate baselines, MixRL at 43.4 for 9B and GRPO at 37.1 for 4B, by 3.7 points at both scales. Compared with GRPO, the gains are 4.8 and 3.8 points on mathematical reasoning and 4.0 and 3.6 points on multi-hop question answering at 9B and 4B, respectively. On the held-out Bamboogle benchmark, BridgeRL obtains pass@2/pass@4 scores of 54.4/60.0 at 9B and 46.4/48.0 at 4B, outperforming the strongest baselines at both sampling budgets and model scales.

Our contributions are:

- We show that GRPO and the centered MaxRL estimator share the same group update direction and differ only in scalar magnitude.
- We introduce an ESS-calibrated actor coefficient that explicitly interpolates between GRPO and MaxRL, with a group-adaptive reward-gain and effective-sample-size criterion selecting the target through a stable grid-search implementation.
- We evaluate the method across six in-domain reasoning benchmarks and an out-of-distribution general QA benchmark at two model scales; a fixed-weight ablation and the Bamboogle results validate adaptation and transfer.

2 Background

2.1 GRPO and MaxRL

GRPO and MaxRL offer complementary ways to convert outcome feedback into policy updates. GRPO provides a practical critic-free update whose within-group standardization yields a normalized, comparatively conservative learning signal (Shao et al. 2024; Yu et al. 2025; Guo et al. 2025). Its limitation is objective-level: collecting more responses improves estimation but leaves the same local reward-linear proxy rather than moving it toward likelihood maximization. MaxRL instead uses rollout compute to form a progressively closer proxy to the maximum-likelihood objective, treating verified responses more like supervised targets (Tajwar et al. 2026). This improves alignment with the desired likelihood-seeking objective, but the resulting update can become excessively sharp and depend on very few trajectories when high-reward responses are sparse. The two methods therefore expose a central tradeoff between conservative normalization and concentrated likelihood seeking.

When verified responses are sparse, GRPO still spreads learning pressure across the sampled group after standardization, whereas MaxRL places substantially more pressure on the few successful trajectories. The latter is closer to likelihood-seeking behavior, but its sharper weighting can make optimization depend heavily on a small portion of the group.

2.2 Listwise Target Projection

Listwise Policy Optimization (LPO) interprets group-based policy gradients as approximate projections toward a reward-induced distribution on the simplex of sampled responses (Qu et al. 2026). It makes this geometry explicit by first constructing a Gibbs target with a temperature-controlled sharpness and then projecting the policy toward that target with forward or reverse KL divergence. The forward-KL variant yields bounded, zero-sum, and self-correcting listwise coefficients, illustrating the stability benefits of exact simplex projection.

LPO's principal practical limitation is its reliance on a prescribed target temperature. The temperature controls the exponential reward tilt, but the resulting concentration also depends on the reward profile of the sampled group. A single fixed value can therefore be too conservative for one group and too concentrated for another, and must be tuned jointly with the projection direction. The temperature sweeps in Appendix Tables 4 and 5 exhibit this task- and metric-dependent behavior. Moreover, the forward-KL coefficient is the raw probability gap between the policy and target, so making the target conservative also reduces the coefficient magnitude; target sharpness and update scale are therefore coupled. These observations motivate an adaptive coefficient that selects sharpness from the current group while preserving GRPO scale in the conservative regime. Our formulation below develops that coefficient from the GRPO and MaxRL endpoints.

3 Preliminaries: GRPO and MaxRL

3.1 Setup

In the following search setting, the reward is binary for convenience. For a prompt x, let

$$p = p_\theta(x) = \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[R(x,y)] \tag{1}$$

be the true policy success probability. Sample K responses $y_{1:K}$ independently from $\pi_\theta(\cdot \mid x)$ and let

$$R_i = R(x,y_i) \in \{0,1\}, \qquad S_i = \nabla_\theta \log \pi_\theta(y_i \mid x). \tag{2}$$

Define the number of successes and empirical success rate as

$$C := \sum_{i=1}^{K} R_i, \qquad \widehat{p} := \frac{C}{K}. \tag{3}$$

For standard on-policy groups, the sampling distribution is uniform:

$$P_i := \frac{1}{K}, \qquad \sum_{i=1}^{K} P_i = 1. \tag{4}$$

The bridge derivation also permits non-uniform weights satisfying $P_i > 0$. For any group statistic $f = (f_1, \ldots, f_K)$, define expectation under P by

$$\mathbb{E}_P[f] := \sum_{i=1}^{K} P_i f_i. \tag{5}$$

The reward mean and variance are

$$\mu_R = \mathbb{E}_P[R], \qquad \sigma_R^2 = \mathbb{E}_P[(R-\mu_R)^2], \tag{6}$$

and, for numerical stabilizer $\epsilon > 0$,

$$z_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}. \tag{7}$$

By construction, the standardized rewards are centered:

$$\mathbb{E}_P[z] = 0. \tag{8}$$

3.2 GRPO Formulation

For a uniform binary-reward group, the reward moments reduce to

$$\mu_R = \widehat{p}, \qquad \sigma_R^2 = \widehat{p}(1-\widehat{p}). \tag{9}$$

Ignoring ϵ when $0 < \widehat{p} < 1$, GRPO therefore assigns

$$A_i^{\mathrm{GRPO}} = \begin{cases} \sqrt{(1-\widehat{p})/\widehat{p}}, & R_i = 1, \\ -\sqrt{\widehat{p}/(1-\widehat{p})}, & R_i = 0. \end{cases} \tag{10}$$

Thus standardization already strengthens a rare observed success. Its functional dependence on $\widehat{p}$ is nevertheless fixed.

3.3 MaxRL Formulation

Under this binary search specialization, MaxRL uses rollout compute to approach the success log-likelihood; Appendix A.1 derives its series construction and endpoint limits. To permit the same comparison for scalar rewards, let r_0 be a fixed reward floor and define

$$U_i := R_i - r_0, \qquad \bar{U} := \mathbb{E}_P[U] = \mu_R - r_0, \qquad \bar{U} > 0. \tag{11}$$

The sampled MaxRL-style coefficient is

$$A_i^{\mathrm{MaxRL}} := \frac{U_i - \bar{U}}{\bar{U} + \epsilon} = \frac{R_i - \mu_R}{\mu_R - r_0 + \epsilon}. \tag{12}$$

For binary rewards with $r_0 = 0$ and the stabilizer omitted at the endpoint, this reduces to $R_i/\widehat{p} - 1$ when $\widehat{p} > 0$. Appendix A.1 derives the corresponding score-function form and shows that its centered coefficient preserves the maximum-likelihood gradient. For a sampled group with at least one success, replacing p by $\widehat{p}$ and the expectation by the weighted group average gives the self-normalized, variance-reduced coefficient

$$A_i^{\mathrm{MaxRL}} := \frac{R_i}{\widehat{p}} - 1, \qquad C > 0. \tag{13}$$

It is exactly centered within the sampled group:

$$\sum_{i=1}^{K} P_i A_i^{\mathrm{MaxRL}} = \frac{\sum_i P_i R_i}{\widehat{p}} - \sum_i P_i = 0. \tag{14}$$

The resulting group update is

$$g_{\mathrm{MaxRL}} = \begin{cases} \displaystyle\sum_{i=1}^{K} P_i A_i^{\mathrm{MaxRL}} S_i, & C > 0, \\ 0, & C = 0. \end{cases} \tag{15}$$

For a non-degenerate group, the successful and unsuccessful coefficients are

$$A_+^{\mathrm{MaxRL}} = \frac{1-\widehat{p}}{\widehat{p}}, \qquad A_-^{\mathrm{MaxRL}} = -1, \qquad 0 < \widehat{p} < 1. \tag{16}$$

3.4 Shared Direction and Magnitude Difference

The two estimators can be compared before introducing an adaptive selector. Both coefficients are centered by the same group mean, so their difference is a scalar multiplier applied to one reward direction.

Proposition 1 (Shared update direction)**.** For any group with $\bar{U} > 0$, the GRPO and scalar-reward MaxRL coefficients satisfy

$$\begin{aligned} A_i^{\mathrm{GRPO}} &= (R_i - \mu_R)\frac{1}{\sigma_R + \epsilon}, \\ A_i^{\mathrm{MaxRL}} &= (R_i - \mu_R)\frac{1}{\mu_R - r_0 + \epsilon}. \end{aligned} \tag{17}$$

Consequently, their group updates are collinear:

$$\begin{aligned} d_R &:= \sum_i P_i (R_i - \mu_R) S_i, \\ g_{\mathrm{GRPO}} &= \frac{1}{\sigma_R + \epsilon}\, d_R, \qquad g_{\mathrm{MaxRL}} = \frac{1}{\mu_R - r_0 + \epsilon}\, d_R. \end{aligned} \tag{18}$$

Thus the methods differ in update magnitude, not in the centered reward direction used by the group update.

4 Method

The preceding proposition identifies update magnitude as the controllable difference between the two endpoints. We retain exponential reward reweighting to search over candidate targets, but calibrate the actor interpolation from the selected target's concentration rather than from raw inverse temperature. The resulting coefficient remains an explicit convex interpolation between the two endpoint estimators.

4.1 Exponential Reward Reweighting

The standardized GRPO statistic z_i provides a centered and scale-normalized ordering of the responses. Introduce a nonnegative reweighting strength a, where larger values favor responses with larger standardized reward. The corresponding tilted distribution is

$$q_i(a) = \frac{P_i \exp(a z_i)}{Z(a)}, \qquad Z(a) = \sum_j P_j \exp(a z_j), \tag{19}$$

where the normalizer $Z(a)$ keeps $q(a)$ on the response simplex. At zero strength, the tilted and sampling distributions coincide; increasing the strength moves progressively more mass toward higher-reward responses. The movement relative to the sampling distribution is represented by the density ratio

$$u_i(a) = \frac{q_i(a)}{P_i} = \frac{\exp(a z_i)}{Z(a)}. \tag{20}$$

For any statistic $f = (f_1, \ldots, f_K)$, define expectation under the tilted distribution by

$$\mathbb{E}_{q(a)}[f] := \sum_{i=1}^{K} q_i(a) f_i. \tag{21}$$

For a uniform on-policy group, we define the conventional effective sample size and its group-normalized form together.

$$\begin{aligned} \mathrm{ESS}(q(a)) &:= \frac{1}{\sum_i q_i(a)^2}, \\ \eta(a) &:= \frac{\mathrm{ESS}(q(a))}{K} = \frac{1}{K \sum_i q_i(a)^2}. \end{aligned} \tag{22}$$



Figure 1: Coefficient illustration for $K = 8$ responses with one verified response. GRPO assigns approximately 2.65 to the verified response and -0.38 to each failure, while MaxRL assigns 7 and -1. The BridgeRL panel uses an illustrative adaptive weight of 0.65, yielding 4.17 and -0.60. All three updates share the same centered reward direction, while BridgeRL selects an intermediate magnitude.

If the target distributes its mass equally over m responses, then $\eta(a) = m/K$; it equals one when the full group is retained and decreases toward $1/K$ when the target concentrates on one response. Appendix A.4 gives the density-ratio extension for non-uniform sampling weights and its partition-function form.

4.2 Adaptive Strength by Reward Gain and ESS

Having quantified target support, we next select the tilted target before constructing the actor coefficient. Increasing a makes the distribution more reward-seeking, but it also concentrates weight on fewer sampled responses. The selector therefore needs one quantity for improvement and another for statistical support.

The natural improvement measure is the change in expected reward caused by replacing the sampling distribution with the tilted distribution. Subtracting the original expectation isolates the benefit of reweighting and ensures that the gain is zero at zero reweighting strength:

$$G(a) = \mathbb{E}_{q(a)}[R] - \mathbb{E}_P[R], \tag{23}$$

Reward gain measures the benefit of moving toward high-reward responses, while relative ESS discounts gains supported by too few samples. Their product gives a support-adjusted reward gain without introducing an additional penalty coefficient:

$$a^\star \in \arg\max_{a \geq 0} \mathcal{M}(a), \qquad \mathcal{M}(a) = G(a)\eta(a). \tag{24}$$

The selected inverse temperature determines the target, after which the target's relative ESS determines the interpolation. The selector score is not multiplied into the actor coefficient, allowing inverse temperature to search the target family without treating its raw value as target concentration.

Across the nonnegative search domain, the two selector components obey

$$G'(a) \geq 0, \qquad \eta'(a) \leq 0. \tag{25}$$

The proof is given in Appendix A.5.

Equation 24 consequently searches between two opposed monotone effects. A clean high-reward subset can justify stronger reweighting, whereas a diffuse reward profile can favor a broader update.

4.3 ESS-Calibrated GRPO–MaxRL Interpolation

The selected strength $a^\star$ identifies a target, but the numerical inverse temperature does not directly quantify how concentrated that target is. We therefore use its relative ESS to calibrate the actor interpolation. Define the maximal-reward set and its sampling mass by

$$\mathcal{I}_{\max} := \{i : R_i = \max_j R_j\}, \qquad p_{\max} := \sum_{i \in \mathcal{I}_{\max}} P_i. \tag{26}$$

The relative ESS spans exactly the support interval induced by the group:

$$\eta(0) = 1, \qquad \lim_{a \to \infty} \eta(a) = p_{\max}. \tag{27}$$

The derivation is deferred to Appendix A.4. For a non-degenerate reward group, normalizing this interval gives the interpolation weight

$$\lambda_\eta(a) := \frac{\eta(a) - p_{\max}}{1 - p_{\max}} \in [0, 1]. \tag{28}$$

The complementary MaxRL weight satisfies

$$1 - \lambda_\eta(a) = \frac{1 - \eta(a)}{1 - p_{\max}}.$$

Thus, it equals the fraction of the group's attainable relative-ESS reduction realized by the target $q(a)$. Evaluating this weight at the selected target gives the final actor advantage:

$$\begin{aligned} \lambda^\star &:= \frac{\eta(a^\star) - p_{\max}}{1 - p_{\max}}, \\ A_i^{\text{new}} &:= \lambda^\star A_i^{\text{GRPO}} + \big(1 - \lambda^\star\big) A_i^{\text{MaxRL}}. \end{aligned} \tag{29}$$

Algorithm 1 summarizes the complete per-group computation.

Algorithm 1 Adaptive Bridge Coefficients for One Group

Require: Rewards $R_{1:K}$, weights $P_{1:K}$, reward floor r_0, search parameters N, $\Delta_{\max}$, J, and ϵ
Compute μ_R, σ_R, $z_{1:K}$, and $\bar{U} = \mu_R - r_0$
Compute $A_i^{\mathrm{GRPO}} = (R_i - \mu_R)/(\sigma_R + \epsilon)$
Compute $A_i^{\mathrm{MaxRL}} = (R_i - \mu_R)/(\bar{U} + \epsilon)$
$\mathcal{I}_{\max} \leftarrow \{i : R_i = \max_j R_j\}$ and $p_{\max} \leftarrow \sum_{i \in \mathcal{I}_{\max}} P_i$
Construct the group-adaptive grid $\mathcal{A}$ using Equations 30–32
for $a \in \mathcal{A}$ **do**
    Compute $\log Z(a)$ and $\log Z(2a)$ by log-sum-exp
    Form $q(a)$, $G(a)$, $\eta(a)$, and $\mathcal{M}(a)$
end for
$a^\star \leftarrow \arg\max_{a \in \mathcal{A}} \mathcal{M}(a)$, breaking ties toward smaller a
Refine $a^\star$ for J golden-section steps between adjacent grid points
$\lambda^\star \leftarrow (\eta(a^\star) - p_{\max})/(1 - p_{\max})$
$A_i^{\mathrm{new}} \leftarrow \lambda^\star A_i^{\mathrm{GRPO}} + (1 - \lambda^\star) A_i^{\mathrm{MaxRL}}$
return $A_{1:K}^{\mathrm{new}}$, $a^\star$, $\lambda^\star$

4.4 Practical Computation

The selector operates on the standardized rewards $z_{1:K}$ from Equation 7. For a non-degenerate group, first define their span:

$$\Delta_z := \max_i z_i - \min_i z_i. \tag{30}$$

Given a prescribed maximum tilted-logit difference $\Delta_{\max}$, the group-specific search limit is

$$a_{\max} := \frac{\Delta_{\max}}{\Delta_z}, \qquad a_{\max}\Delta_z = \Delta_{\max}. \tag{31}$$

Let $\mathcal{A}$ denote the candidate reweighting strengths. For N candidates, we use the quadratic grid

$$\mathcal{A} := \left\{ a_{\max} \left(\frac{n-1}{N-1} \right)^2 \,\middle|\, n = 1, \ldots, N \right\}. \tag{32}$$

This grid allocates more candidates near the conservative endpoint. We select its best candidate and refine the interval between adjacent candidates using golden-section search. Groups with fewer than two responses or reward standard deviation below ϵ bypass adaptive selection and retain zero coefficients. All log-partition computations use log-sum-exp, and the selected strength and coefficients are detached before forming the policy loss.

The coefficients can replace GRPO advantages inside the same clipped token-level surrogate. This preserves the existing optimizer, reference-model penalty, and masking rules; only the group-level coefficient changes.

5 Experiments

We evaluate BridgeRL on two categories of reasoning tasks that test different generation patterns: search-based multi-hop question answering, where the policy must reason and call a retrieval tool to compose evidence, and mathematical reasoning, where the policy produces self-contained solutions scored by an outcome verifier. We first describe the benchmarks, baselines, implementation details, and reporting conventions in Section 5.1, and then present matched comparisons at 9B and 4B scales in Section 5.2. Section 6 isolates the contribution of adaptive interpolation and evaluates out-of-distribution transfer, while the Appendix reports broader fixed-temperature and fixed-interpolation sweeps.

5.1 Experimental Setup

Training datasets. For multi-hop reasoning, we use the first 5,000 training examples from each of 2WikiMultiHopQA (Ho et al. 2020), MuSiQue (Trivedi et al. 2022), and HotpotQA (Yang et al. 2018). During rollout, the policy uses the wikipedia Python package as its search tool (Goldsmith 2014). For mathematical reasoning, we sample 4,500 examples from BytedTsinghua-SIA/DAPO-Math-17k, the training collection released with DAPO (Yu et al. 2025).

Evaluation datasets. Mathematical reasoning is evaluated on the Level-5 subset of DigitalLearningGmbH/MATH-lighteval, AI-MO/aimo-validation-amc, and AI-MO/NuminaMath-CoT. Multi-hop reasoning is evaluated on 2WikiMultiHopQA, MuSiQue, HotpotQA, and Bamboogle (Press et al. 2023). Unless otherwise specified, each evaluation set contains 500 examples. Bamboogle uses all 125 available examples, and AI-MO/aimo-validation-amc uses all 83 available examples.

Baselines and backbone. We compare GRPO, MaxRL, MixRL, the forward and backward LPO projections, and adaptive BridgeRL. Reported runs use Qwen3.5 backbones at 9B and 4B scales (Yang et al. 2025). All methods share the same distributed GRPO training stack, evaluation prompts, sample counts, and verifier rules; they differ only in the group-level coefficient used to optimize the policy. GRPO and MaxRL provide the two endpoint estimators. In the main comparison, MixRL fixes the GRPO weight to $\lambda = 0.5$ for every group, while both LPO projections fix the inverse temperature to $a = 1$. Appendix Tables 4–6 vary these settings on GRPO-4B to isolate the effects of globally fixing a and λ.

Implementation details. The GRPO rollout group size is $n = 8$, denoted by $K = 8$ in our derivations, and the same group size is used for every matched method. The lower and upper clipping bounds are both set to 0.2. The KL penalty coefficient is 0.001, the batch size is 32, and the learning rate is 3×10^{-6}. We train all models for three epochs to ensure full convergence and report final-checkpoint validation results in Tables 1–2. For search tasks, one turn comprises one reasoning segment followed by one tool call.

Following Section 4.4, we set the maximum tilted-logit difference to $\Delta_{\max} = 20$, evaluate $N = 64$ quadratic-grid candidates per valid group, and apply $J = 16$ golden-section refinement steps. Experiments are conducted on clusters of NVIDIA H200 GPUs with 144GB memory. The implementation is built on PyTorch (Paszke et al. 2019) and uses veRL for distributed reinforcement-learning training (Sheng et al. 2024).

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
GRPO	pass@k	78.8 / 85.6 / 89.6	61.4 / 68.7 / 72.3	38.2 / 45.4 / 50.4	29.2 / 36.6 / 42.4	9.2 / 11.6 / 14.2	25.0 / 30.0 / 31.0	**43.2** / 49.6 / 53.6
	avg@k	78.8 / 77.8 / 78.6	61.4 / 62.4 / 60.7	38.2 / 37.2 / 36.7	29.2 / 29.1 / 29.3	9.2 / 9.3 / 9.2	25.0 / 23.4 / 22.9	**43.2** / 41.5 / 40.7
LPO forward	pass@k	74.2 / 85.0 / 91.2	62.7 / 67.5 / 72.3	35.2 / 42.8 / 48.2	27.6 / 34.6 / 38.2	7.4 / 13.2 / 15.8	21.4 / 29.8 / 30.2	38.4 / 47.2 / 55.2
	avg@k	74.2 / 75.6 / 74.3	62.7 / 62.4 / 60.7	35.2 / 35.3 / 36.8	27.6 / 29.1 / 27.7	7.4 / 7.4 / 9.1	21.4 / 23.4 / 21.3	38.4 / 38.2 / 40.7
LPO backward	pass@k	74.4 / 82.8 / 85.2	59.0 / 65.1 / 68.7	38.2 / 45.6 / 46.8	28.6 / 37.0 / 40.8	8.0 / 11.0 / 11.8	19.4 / 30.4 / 32.2	39.2 / 44.8 / 54.4
	avg@k	74.4 / 77.8 / 77.8	59.0 / 58.6 / 60.7	38.2 / 37.2 / 36.7	28.6 / 28.5 / 29.3	8.0 / 9.3 / 9.8	19.4 / 19.5 / 19.5	39.2 / 40.5 / 40.5
MaxRL	pass@k	75.6 / 83.4 / 87.8	54.2 / 63.9 / 68.7	30.8 / 43.4 / 50.6	31.8 / 37.6 / 42.0	10.8 / 11.0 / 13.0	**25.4** / 28.6 / 31.2	**43.2** / 47.2 / 52.8
	avg@k	75.6 / 75.6 / 77.8	54.2 / 54.4 / 54.4	30.8 / 30.8 / 30.8	31.8 / 29.1 / 29.3	10.8 / 9.3 / 9.8	**25.4** / 23.4 / 22.9	**43.2** / 41.5 / 40.7
MixRL	pass@k	81.2 / 84.4 / 93.0	62.7 / 67.5 / 71.1	**40.6** / 44.8 / 52.0	**32.6** / 33.4 / **43.4**	10.8 / 13.6 / 17.4	22.4 / 29.2 / 32.2	40.8 / 51.2 / 56.8
	avg@k	81.2 / 77.7 / 77.7	62.7 / 63.9 / 65.0	**40.6** / 37.2 / 36.8	**32.6** / 29.1 / 29.3	10.8 / 9.3 / 9.1	22.4 / 23.4 / 22.9	40.8 / 40.5 / 40.5
BridgeRL	pass@k	**82.6 / 91.4 / 95.2**	**68.7 / 75.9 / 78.3**	38.2 / **50.0 / 54.6**	31.6 / **38.2** / 42.4	**14.6 / 18.2 / 20.2**	25.2 / **35.4 / 38.6**	**43.2 / 54.4 / 60.0**
	avg@k	**82.6 / 82.1 / 83.0**	**68.7 / 67.8 / 66.8**	38.2 / **41.6 / 42.0**	31.6 / **35.0 / 34.8**	**14.6 / 16.2 / 13.6**	25.2 / **25.3 / 27.8**	**43.2 / 45.1 / 44.5**

Table 1: Per-task 9B validation reward values (%) from the final checkpoint after three training epochs. Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
GRPO	pass@k	70.2 / 79.2 / 84.2	48.2 / 59.0 / 65.1	35.8 / 45.4 / 52.4	26.8 / 31.0 / 33.8	6.0 / 7.0 / 10.0	**21.0** / 21.0 / 24.0	**33.6** / 39.2 / 43.2
	avg@k	70.2 / 69.6 / 72.0	48.2 / 48.5 / 47.6	35.8 / 33.9 / 35.3	26.8 / 25.2 / 24.8	6.0 / 6.8 / 6.8	**21.0** / 18.0 / 19.9	**33.6** / 31.0 / 32.6
LPO forward	pass@k	64.8 / 78.6 / 89.0	48.2 / 59.0 / 66.3	35.4 / 44.8 / 47.4	22.6 / 31.2 / 33.4	6.8 / 7.6 / 10.4	15.8 / 22.8 / 25.4	32.0 / 40.0 / 46.4
	avg@k	64.8 / 69.3 / 69.9	48.2 / 47.6 / 47.6	35.4 / 32.4 / 35.3	22.6 / 25.7 / 24.8	6.8 / 6.8 / 6.8	15.8 / 15.8 / 15.8	32.0 / 30.9 / **32.8**
LPO backward	pass@k	64.8 / 75.6 / 83.2	47.0 / 55.4 / 63.9	**37.2** / 48.6 / 49.8	**28.8** / 29.4 / 35.6	5.2 / 8.0 / 9.0	17.6 / 19.2 / 26.8	28.0 / 33.6 / 43.2
	avg@k	64.8 / 69.8 / 69.8	47.0 / 47.6 / 47.6	**37.2** / 33.9 / 35.3	**28.8** / 24.6 / 28.5	5.2 / 7.0 / 7.2	17.6 / 17.7 / 17.7	28.0 / 31.0 / 30.8
MaxRL	pass@k	70.2 / 77.2 / 83.6	41.0 / 50.6 / 63.9	29.2 / 43.4 / 52.4	28.0 / 31.2 / **36.6**	6.4 / 8.8 / 11.8	20.2 / 22.2 / 23.0	31.2 / 36.0 / 40.0
	avg@k	70.2 / 68.7 / 69.8	41.0 / 40.8 / 40.8	29.2 / 32.4 / 29.1	28.0 / 25.5 / **28.6**	6.4 / 6.4 / 6.9	20.2 / 18.0 / 19.9	31.2 / 30.9 / 30.8
MixRL	pass@k	74.8 / 76.2 / 90.2	48.2 / 55.4 / 67.5	35.0 / 44.2 / 51.0	27.0 / 29.0 / 34.6	4.2 / 7.6 / 11.8	19.8 / 21.4 / 28.2	**33.6** / 44.0 / 44.0
	avg@k	74.8 / 69.0 / 69.8	48.2 / 48.6 / 47.9	35.0 / 33.9 / 35.3	27.0 / 24.9 / 24.8	4.2 / 4.2 / 4.2	19.8 / 18.0 / 19.9	**33.6** / 30.9 / 30.8
BridgeRL	pass@k	**77.4 / 81.8 / 92.2**	**51.8 / 62.7 / 72.3**	34.4 / **50.8 / 54.4**	**28.8 / 33.8 / 36.6**	**10.2 / 13.0 / 17.8**	**21.0 / 29.6 / 30.0**	32.8 / **46.4 / 48.0**
	avg@k	**77.4 / 75.6 / 75.6**	**51.8 / 51.1 / 51.7**	34.4 / **37.7 / 36.3**	**28.8 / 27.7** / 28.5	**10.2 / 8.5 / 8.3**	**21.0 / 24.0 / 22.1**	32.8 / **31.7** / 32.7

Table 2: Per-task 4B validation reward values (%) from the final checkpoint after three training epochs. Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Metrics and reporting. We report verifier reward after task-specific answer normalization. For each prompt, pass@k records whether at least one of k sampled responses receives a positive verifier reward, while avg@k averages the verifier reward across those k responses. Tables 1 and 2 report both metrics at $k \in \{1, 2, 4\}$ for every benchmark. Bold entries mark the best method for each task, metric, and sampling budget. Aggregate comparisons average the six reported metrics across the six in-domain benchmarks, excluding Bamboogle, which is reserved for out-of-distribution analysis in Section 6.2.

5.2 Main Results

Tables 1 and 2 compare BridgeRL with the five baselines at both model scales. Across the six in-domain benchmarks and all six reported metrics, BridgeRL obtains averages of 47.1 at 9B and 40.8 at 4B. The strongest aggregate competitors are MixRL at 43.4 for 9B and GRPO at 37.1 for 4B, leaving margins of 3.7 points at both scales. On mathematics, BridgeRL averages 67.1 and 59.4, compared with GRPO's 62.3 and 55.6, for gains of 4.8 and 3.8 points. On in-domain multi-hop QA, it averages 27.1 and 22.2, exceeding GRPO's 23.1 and 18.7 by 4.0 and 3.6 points at 9B and 4B, respectively.

The improvements are consistent across tasks and sampling budgets. Against the strongest competing value in each task-metric comparison, BridgeRL improves the average by 3.1 points at 9B and 2.1 points at 4B. It is strictly best in 33 of 42 comparisons at 9B and 31 of 42 at 4B, and ties the best result in another 2 and 5 comparisons, respectively. BridgeRL achieves the best pass@2 and avg@2 on all seven tasks at both scales. At pass@4, it leads six tasks at 9B; at 4B, it leads six and ties MaxRL on 2WikiMultiHopQA. The broader GRPO-4B sweeps in Appendix Tables 4–6 show that changing either the LPO temperature or the fixed interpolation weight redistributes gains among tasks rather than producing one uniformly dominant setting.

Coefficient rule	9B	4B
Reward gain only (MaxRL)	40.7	35.6
ESS only (GRPO)	42.7	37.1
Fixed $\lambda = 0.5$ (MixRL)	43.4	37.1
Adaptive BridgeRL	**47.1**	**40.8**

Table 3: Six-metric average reward across the six in-domain benchmarks. Reward-gain-only and ESS-only selection reduce to the two endpoint estimators, while MixRL fixes the interpolation weight for every group.

6 Analysis

6.1 Ablation Study

A selector based only on reward gain or only on ESS does not produce a genuinely intermediate method. Because reward gain is nondecreasing over the search grid, gain-only selection chooses the largest available reweighting strength and approaches the MaxRL endpoint. Because relative ESS is nonincreasing, ESS-only selection chooses zero reweighting strength and exactly recovers the GRPO endpoint. On the finite grid, these relations are summarized by

$$\begin{aligned} a_{\max} \in \arg\max_{a\in\mathcal{A}} G(a), \quad A^{\mathrm{new}}(a_{\max}) \approx A^{\mathrm{MaxRL}}, \\ 0 \in \arg\max_{a\in\mathcal{A}} \eta(a), \qquad A^{\mathrm{new}}(0) = A^{\mathrm{GRPO}}. \end{aligned} \tag{33}$$

We therefore use fixed interpolation as the informative ablation. Here $\lambda \in [0,1]$ is the GRPO weight, so fixing it also fixes the complementary MaxRL weight for every sampled group, regardless of that group's reward profile or maximal-reward support. The complete fixed-interpolation family is

$$A_i^{\mathrm{MixRL}}(\lambda) = \lambda A_i^{\mathrm{GRPO}} + (1-\lambda) A_i^{\mathrm{MaxRL}}, \qquad \lambda \in [0,1]. \tag{34}$$

The main comparison uses the midpoint, while Appendix Table 6 evaluates three fixed weights.

Table 3 shows that a fixed midpoint is insufficient. At 9B, MixRL improves the in-domain average from 42.7 under GRPO to 43.4, but remains 3.7 points below adaptive BridgeRL. At 4B, the fixed mixture reaches 37.1, matching GRPO to one decimal place, whereas adaptive BridgeRL reaches 40.8 and improves over the fixed mixture by 3.7 points. The broader sweep gives aggregate scores of 36.9, 37.1, and 36.8 for $\lambda = 0.25$, 0.5, and 0.75, respectively. Fixing λ removes the mechanism that responds to group support: the same relative update magnitude is used whether maximal-reward responses are scarce or well supported. No tested global weight closes the gap to per-group adaptation.

Appendix Tables 4–6 provide broader ablations based on GRPO-4B. The two LPO sweeps show how fixing a chooses one reward-tilt strength for all groups under each projection direction. The interpolation sweep shows how fixing λ instead chooses one relative GRPO–MaxRL magnitude for all groups. Their task-dependent optima reinforce the main result: neither one prescribed projection temperature nor one fixed interpolation weight captures every benchmark and sampling budget.

6.2 Out-of-Distribution Evaluation

We use Bamboogle as the out-of-distribution benchmark. It is held out from all training sources and consists of general compositional question answering rather than problems tailored specifically to the training-time search datasets or to mathematical reasoning. It therefore tests whether the learned optimization rule transfers beyond both source distributions while retaining the ability to combine evidence.

At 9B, BridgeRL achieves Bamboogle pass@2 and pass@4 scores of 54.4 and 60.0. These exceed the strongest baselines, MixRL at 51.2 and 56.8, by 3.2 points at both budgets. It also obtains the best avg@2 and avg@4 scores, improving over the strongest baselines by 3.6 and 3.8 points. At 4B, BridgeRL again gives the best pass@2 and pass@4 results, reaching 46.4 and 48.0, with margins of 2.4 and 1.6 points. Its avg@2 score leads by 0.7 points, while its avg@4 score of 32.7 is within 0.1 points of LPO forward. These results show that the multi-sample gains extend to a held-out general QA distribution at both model scales.

7 Conclusion

We introduced BridgeRL, an adaptive sampled-group coefficient between GRPO and MaxRL. A shared-direction derivation shows that the two endpoint estimators differ in scalar magnitude, motivating an explicit convex interpolation. Exponential reward reweighting searches the target family, a reward-gain/ESS criterion selects the target, and normalized target ESS calibrates the actor coefficient relative to the group's maximal-reward support. Across the six in-domain benchmarks, BridgeRL reaches 47.1 at 9B and 40.8 at 4B, improving over the strongest aggregate baseline by 3.7 points at both scales. Relative to GRPO, it gains 4.8 and 3.8 points on mathematics and 4.0 and 3.6 points on multi-hop question answering at the two scales. The fixed-weight and temperature sweeps show that globally prescribed settings redistribute gains across tasks, while the Bamboogle results demonstrate that the multi-sample improvements transfer to a held-out general QA distribution.

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